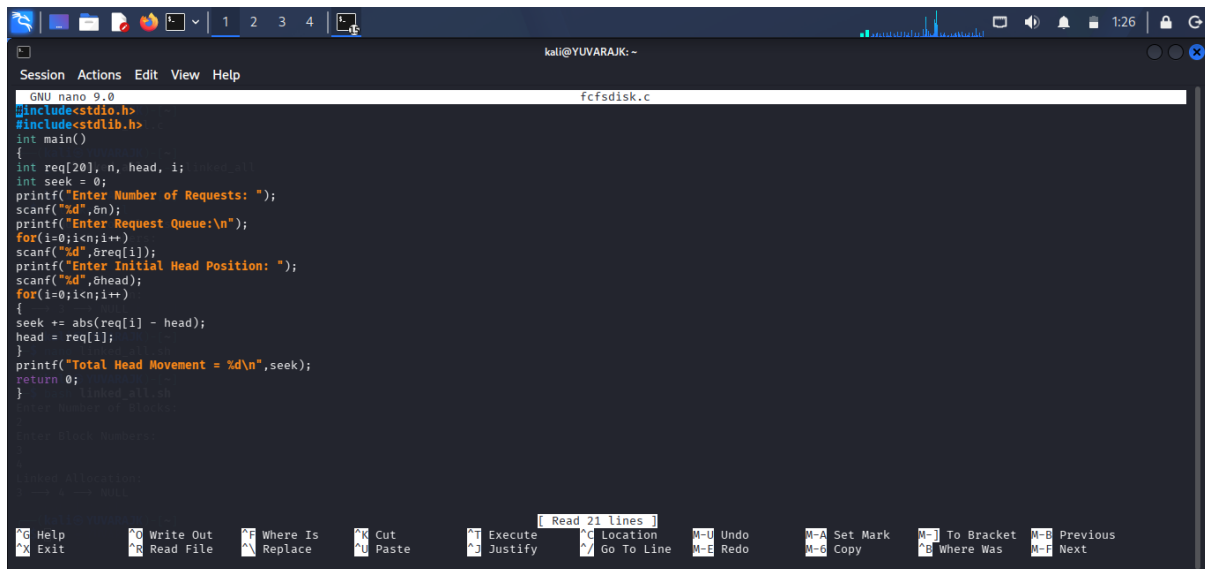
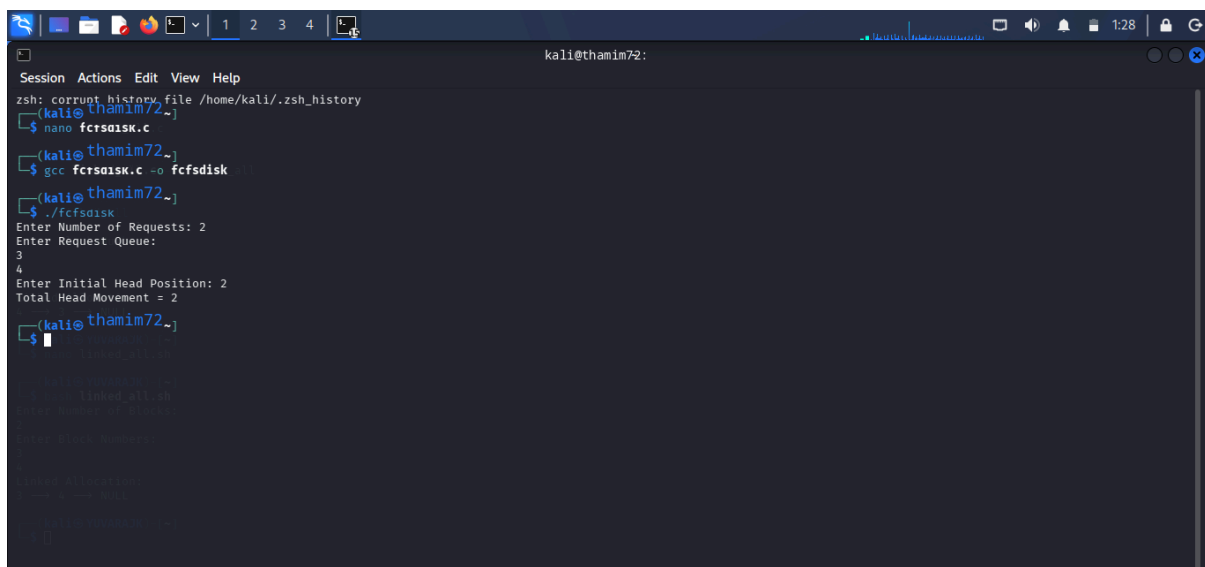


C PROGRAMMING :FCFS DISK



```
GNU nano 9.0 fcfdisk.c
#include<stdio.h>
#include<stdlib.h>
int main()
{
    int req[20], n, head, i;
    int seek = 0;
    printf("Enter Number of Requests: ");
    scanf("%d", &n);
    printf("Enter Request Queue:\n");
    for(i=0; i<n; i++)
    {
        scanf("%d", &req[i]);
    }
    printf("Enter Initial Head Position: ");
    scanf("%d", &head);
    for(i=0; i<n; i++)
    {
        seek += abs(req[i] - head);
        head = req[i];
    }
    printf("Total Head Movement = %d\n", seek);
    return 0;
}
```

OUTPUT :



```
kali@thamim72:~$ nano fcfdisk.c
kali@thamim72:~$ gcc fcfdisk.c -o fcfdisk
kali@thamim72:~$ ./fcfsdisk
Enter Number of Requests: 2
Enter Request Queue:
3
4
Enter Initial Head Position: 2
Total Head Movement = 2
kali@thamim72:~$
```

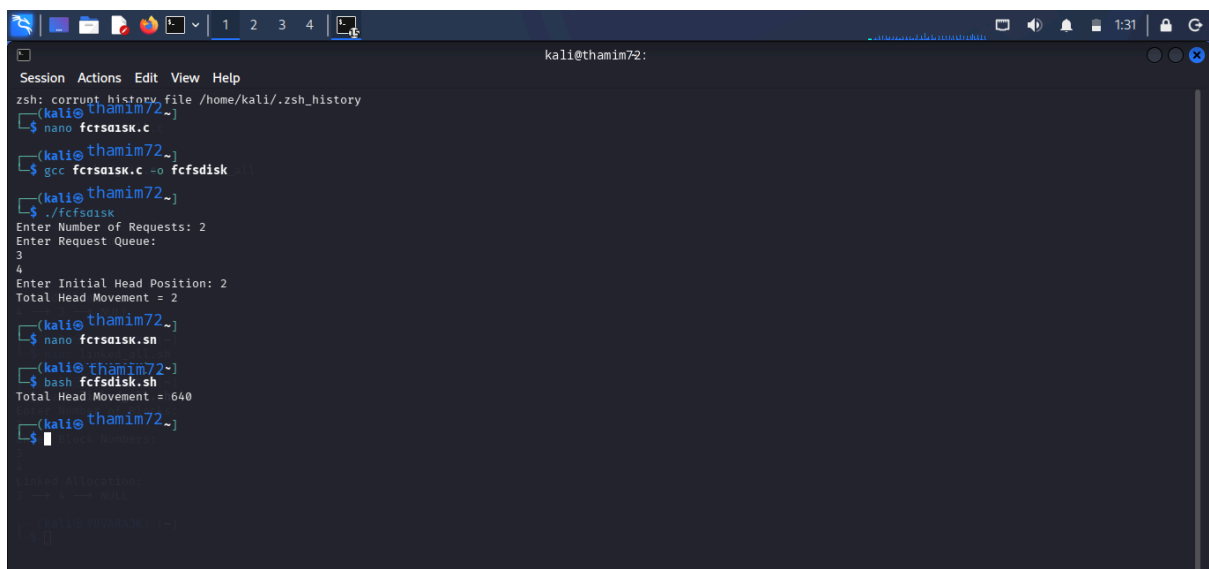
SHELL PROGRAMMING : FCFS DISK



The screenshot shows a terminal window with the nano 9.0 editor open. The file being edited is fcfsdisk.sh. The script is a shell program that implements the First-Come, First-Served (FCFS) disk scheduling algorithm. It takes a queue of requests as input and calculates the total head movement required to service them in order. The script is as follows:

```
#!/bin/bash
queue=(98 183 37 122 14 124 65 67)
head=53
seek=0
for req in "${queue[@]}"; do
    diff=$((req-head))
    if [ $diff -lt 0 ]; then
        then head=$((head-diff))
    fi
    seek=$((seek+diff))
    head=$req
done
echo "Total Head Movement = $seek"
```

OUTPUT :



The screenshot shows a terminal window with the following commands and output:

```
zsh: corrupt history file /home/kali/.zsh_history
kali@thamim72:~$ nano fcfsdisk.c
kali@thamim72:~$ gcc fcfsdisk.c -o fcfsdisk
kali@thamim72:~$ ./fcfsdisk
Enter Number of Requests: 2
Enter Request Queue:
3
4
Enter Initial Head Position: 2
Total Head Movement = 2
kali@thamim72:~$ nano fcfsdisk.sh
kali@thamim72:~$ bash fcfsdisk.sh
Total Head Movement = 640
kali@thamim72:~$
```

C PROGRAMMING : SSTF DISK



```
GNU nano 9.0 stsfdisk.c *
#include<stdio.h>
#include<stdlib.h>
int main() {
    int req[20], visited[20]={0};
    int n, head, i, count=0;
    int seek=0, index, min, distance;
    printf("Enter Number of Requests: ");
    scanf("%d",&n);
    printf("Enter Request Queue:\n");
    for(i=0;i<n;i++){
        scanf("%d",&req[i]);
    }
    printf("Enter Initial Head Position: ");
    scanf("%d",&head);
    while(count<n){
        min=9999;
        for(i=0;i<n;i++){
            if(!visited[i]){
                distance=abs(req[i]-head);
                if(distance<min){
                    min=distance;
                    index=i;
                }
            }
        }
        seek+=min;
        head=req[index];
        visited[index]=1;
        count++;
    }
    printf("Total Head Movement = %d\n",seek);
    return 0;
}
```

OUTPUT :

[illegible]

SHELL PROGRAMMING : SSTF DISK

The screenshot shows a Kali Linux terminal window. The title bar indicates the user is 'kali@YUVARAJIK' in the home directory. The terminal is running a nano editor session for a file named 'sstfdisk.sh'. The script's output is as follows:

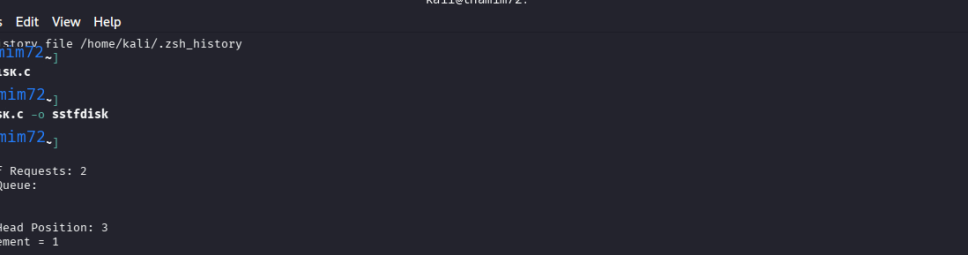
```
GNU nano 9.0 sstfdisk.sh
#!/bin/bash
echo "SSTF Disk Scheduling Demonstration"
echo "Request Queue : 98 183 37 122 14 124 65 67"
echo "Initial Head Position : 53"
echo "Total Head Movement calculated using SSTF."
```

The bottom of the terminal features a command palette with the following options:

- Help
- Write Out
- Where Is
- Cut
- Execute
- Justify
- Location
- Go To Line
- Undo
- Redo
- Set Mark
- Copy
- To Bracket
- Where Was
- Previous
- Next

The 'Read 5 lines' option is currently selected in the palette.

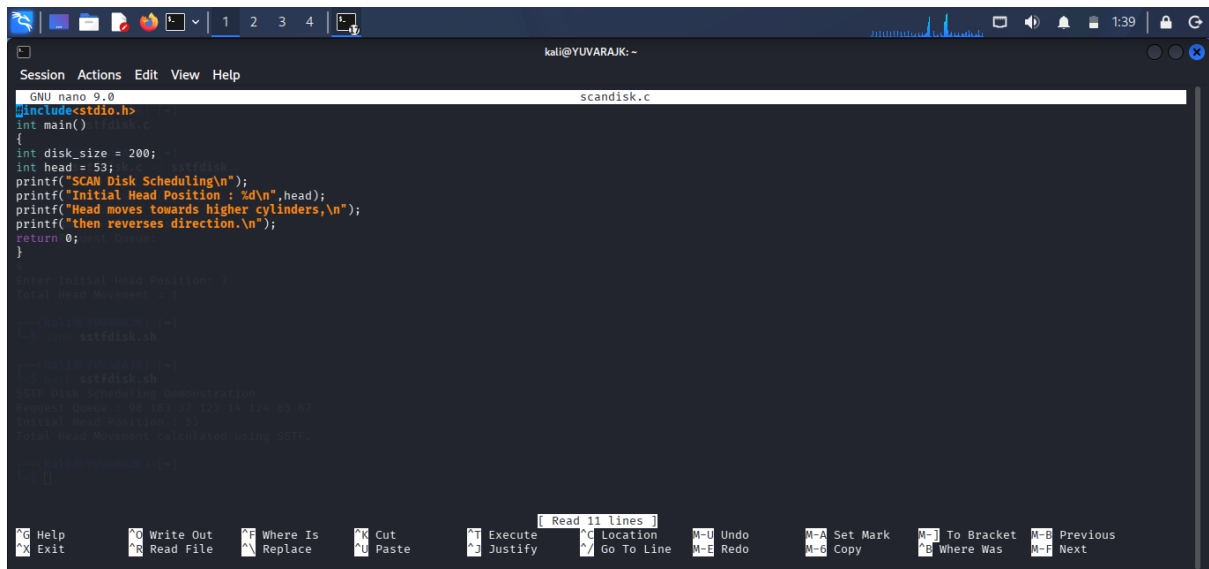
OUTPUT :



The screenshot shows a Kali Linux terminal window with the following content:

```
kali@thamim72:~$ zsh: corrupt history file /home/kali/.zsh_history
(kali@thamim72~)
$ nano sstrodisk.c
(kali@thamim72~)
$ gcc sstrodisk.c -o sstfdisk
(kali@thamim72~)
$ ./sstfdisk
Enter Number of Requests: 2
Enter Request Queue:
3
4
Enter Initial Head Position: 3
Total Head Movement = 1
(kali@thamim72~)
$ nano sstrodisk.sn
(kali@thamim72~)
$ bash sstrodisk.sn
SSTF Disk Scheduling Demonstration
Request Queue : 98 183 37 122 14 124 65 67
Initial Head Position : 53
Total Head Movement calculated using SSTF.
(kali@thamim72~)
$
```

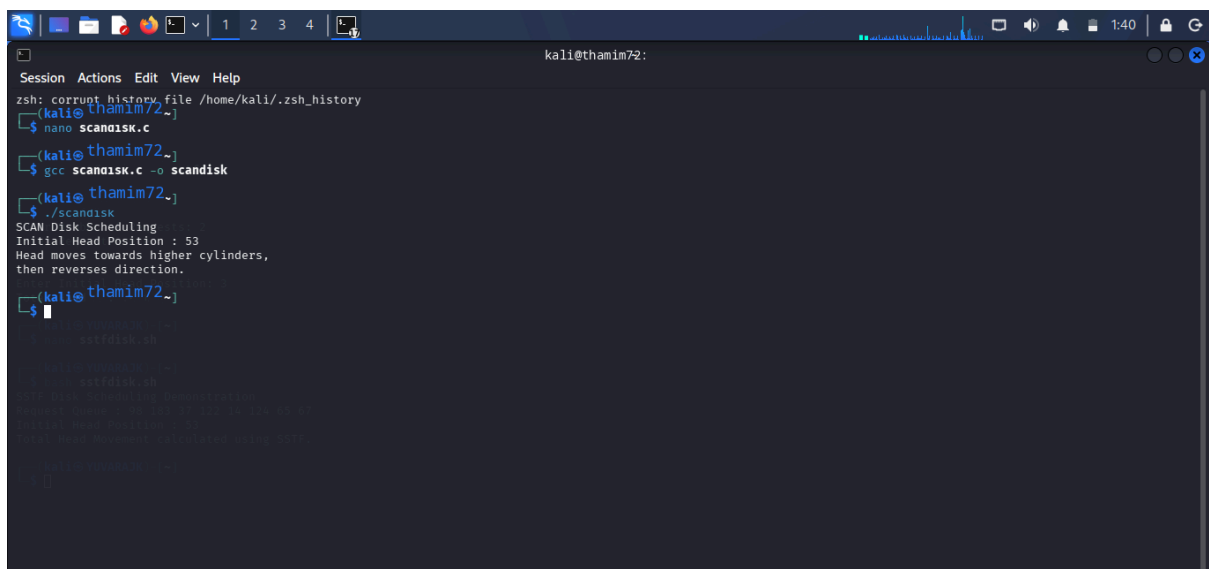
C PROGRAMMING : SCAN DISK



```
GNU nano 9.0 scandisk.c
#include<stdio.h>
int main()
{
    int disk_size = 200;
    int head = 53;
    printf("SCAN Disk Scheduling\n");
    printf("Initial Head Position : %d\n",head);
    printf("Head moves towards higher cylinders,\n");
    printf("then reverses direction.\n");
    return 0;
}
```

The screenshot shows a nano editor window titled 'scandisk.c'. The code defines a disk size of 200 and an initial head position of 53. It prints the disk scheduling algorithm name and the initial head position. The program is intended to simulate the SCAN disk scheduling algorithm, which moves the head towards higher cylinder numbers until it reaches the end of the disk, then reverses direction.

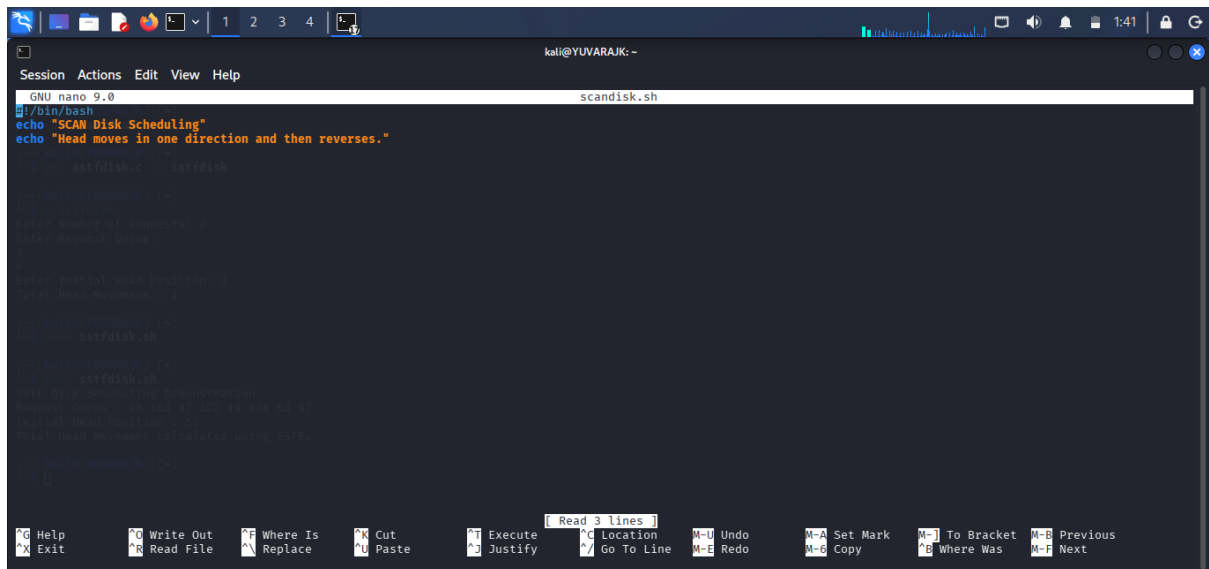
OUTPUT :



```
zsh: current history file /home/kali/.zsh_history
(kali@thamim72:~)
$ nano scandisk.c
(kali@thamim72:~)
$ gcc scandisk.c -o scandisk
(kali@thamim72:~)
$ ./scandisk
SCAN Disk Scheduling
Initial Head Position : 53
Head moves towards higher cylinders,
then reverses direction.
(kali@thamim72:~)
$
```

The screenshot shows a terminal window where the 'scandisk.c' program is compiled and executed. The output of the program is displayed, showing the disk scheduling algorithm name and the initial head position. The program is intended to simulate the SCAN disk scheduling algorithm, which moves the head towards higher cylinder numbers until it reaches the end of the disk, then reverses direction.

SHELL PROGRAMMING : SCAN DISK



The screenshot shows a terminal window with the nano editor open, editing a file named `scandisk.sh`. The script content is as follows:

```
#!/bin/bash
echo "SCAN Disk Scheduling"
echo "Head moves in one direction and then reverses."

# Initial Head Position : 53
# Head moves towards higher cylinders,
# then reverses direction.

# Initial Head Position : 53
# Head moves towards higher cylinders,
# then reverses direction.

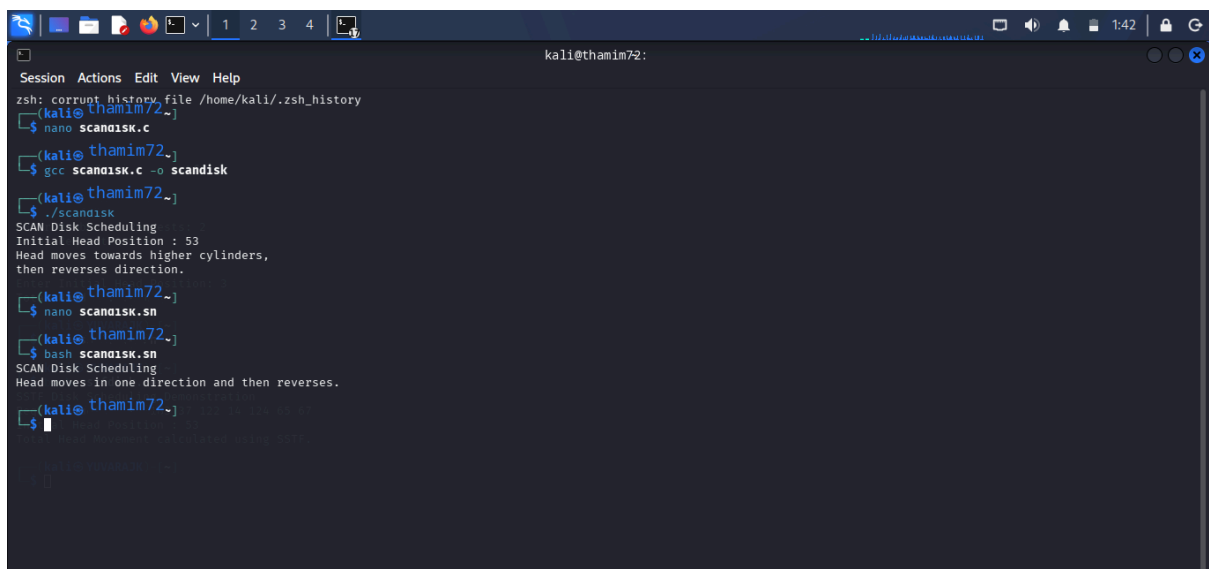
# Initial Head Position : 53
# Head moves towards higher cylinders,
# then reverses direction.

# Initial Head Position : 53
# Head moves towards higher cylinders,
# then reverses direction.

# Initial Head Position : 53
# Head moves towards higher cylinders,
# then reverses direction.

# Initial Head Position : 53
# Head moves towards higher cylinders,
# then reverses direction.
```

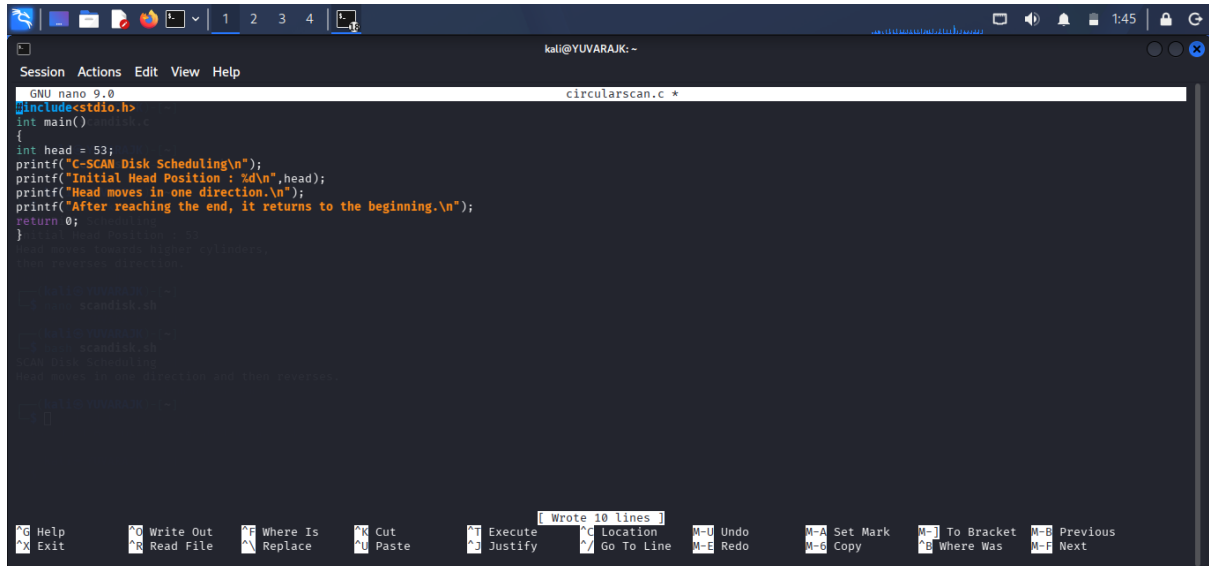
OUTPUT :



The screenshot shows a terminal window with the following commands and output:

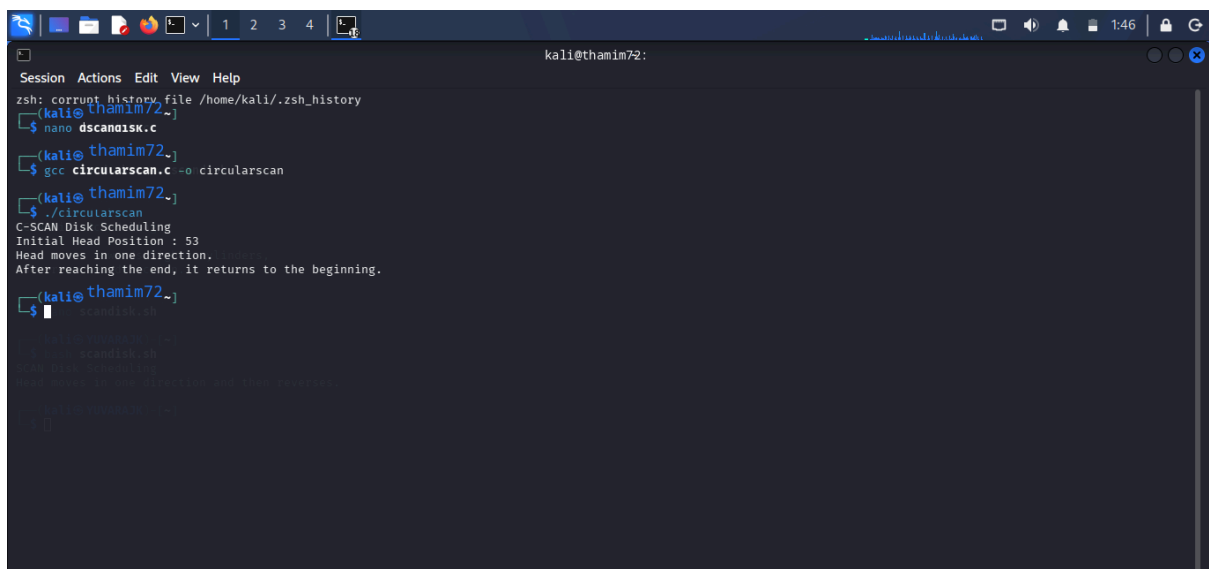
```
zsh: corrupt history file /home/kali/.zsh_history
(kali@thamim72~)
$ nano scandisk.c
(kali@thamim72~)
$ gcc scandisk.c -o scandisk
(kali@thamim72~)
$ ./scandisk
SCAN Disk Scheduling
Initial Head Position : 53
Head moves towards higher cylinders,
then reverses direction.
(kali@thamim72~)
$ nano scandisk.sn
(kali@thamim72~)
$ bash scandisk.sn
SCAN Disk Scheduling
Head moves in one direction and then reverses.
```

C PROGRAMMING : CIRCULAR SCAN



```
GNU nano 9.0 circularscan.c *
#include<stdio.h>
int main()
{
    int head = 53;
    printf("C-SCAN Disk Scheduling\n");
    printf("Initial Head Position : %d\n",head);
    printf("Head moves in one direction.\n");
    printf("After reaching the end, it returns to the beginning.\n");
    return 0;
}
// Initial Head Position : 53
// Head moves in one direction.
// After reaching the end, it returns to the beginning.
// C-SCAN Disk Scheduling
// Head moves in one direction and then reverses.
// After reaching the end, it returns to the beginning.
// 0 1
```

OUTPUT :



```
kali@thamim72:~$ nano dscan01sk.c
kali@thamim72:~$ gcc circularscan.c -o circularscan
kali@thamim72:~$ ./circularscan
C-SCAN Disk Scheduling
Initial Head Position : 53
Head moves in one direction.
After reaching the end, it returns to the beginning.
```

SHELL PROGRAMMING : CIRCULAR SCAN

[illegible]

OUTPUT :

```

kali@thamim72:
Session Actions Edit View Help
zsh: corrupt history file /home/kali/.zsh_history
(kali@thamim72~)
$ nano dscan01sk.c
(kali@thamim72~)
$ gcc circularscan.c -o circularscan
(kali@thamim72~)
$ ./circularscan
C-SCAN Disk Scheduling
Initial Head Position : 53
Head moves in one direction.
After reaching the end, it returns to the beginning.
(kali@thamim72~)
$ nano dscan01sk.sh
(kali@thamim72~)
$ bash circularscan.sh
C-SCAN Disk Scheduling
Head moves in one direction only.
After reaching the end, it returns to the beginning.
(kali@thamim72~)
$
```